Peihua Wang

CONTACT & PROFILES

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RESEARCH INTEREST

Modeling infectious disease and microbial transmission across populations, built environments, and microbial communities

PROFESSIONAL EXPERIENCE

Department of Epidemiology, Mailman School of Public Health, Columbia University, New York, USA

Postdoctoral Research Scientist — Nov 2022–Jun 2025

Department of Mechanical Engineering, The University of Hong Kong, HKSAR

Postdoctoral Fellow — Sept 2021–Nov 2022

EDUCATION

The University of Hong Kong, HKSAR

Ph.D. Mechanical Engineering — Sept 2016–Aug 2021

Thesis: Hitchhiking of microbes and their friendships

Zhejiang University, Hangzhou, China

B.S. Environmental Engineering — Sept 2012–June 2016

HONORS & AWARDS

HK Chief Executive's Commendation for Community Service for outstanding contribution to the fight against COVID-19	2021
HKU-Stanford MedTech Hackathon, 2nd runner up	2019
Goldman Sachs Coding Challenge Award	2019
HKU Philip K H Wong Foundation Postgraduate Fellowship	2016
HKU Y S and Christabel Lung Postgraduate Scholarship	2016
HKU Postgraduate Scholarship	2016

GRANTS

Simulation of microbial transmission cycle in built environment, Procter & Gamble hygiene innovation fund — US$ 419 890, 2021

Role: equivalent to PI (but not eligible in 2021), wrote proposal and ethics application, and supervised students on field experiments and data analyses

MANUSCRIPTS UNDER REVIEW (* indicates corresponding author)

1. **Wang P**, Zhao P, Lim C, Wang H, Lam Z, Liu J, Lee PKH, et al. *Using Lactobacillus strains to simulate multi-pathway fomite-mediated bacterial transfer in households*. Under review in Environment International. 2026.

PUBLICATIONS

1. **Wang P***, Wang X, Pei S, Xu X, Zhang W, Wang Y, Yang W. *Rural-to-urban migrant worker mobility shaped measles epidemics in China*. PLOS Computational Biology. 2026;22(4):e1014182.
2. **Wang P***, Chen J, Zhang W, Wang Y, Yang W. *Modeling the influences of climate conditions on measles transmission in China*. Epidemiology & Infection. 2025;153:e110.
3. Zhao P, **Wang P***, Zhang N, Lam TH, Wang H, Liu J, Li Y. *Experimental tracing and modeling of exogenous microbe dynamics in home environments*. Environmental Science & Technology. 2025;59(50):27474–85.
4. **Wang P**, Tong X, Zhang N, Miao T, Chan Jack PT, Huang H, Lee PKH, et al. *Fomite transmission follows invasion ecology principles*. mSystems. 2022;7(3):e00211–22.
5. **Wang P**, Zhang N, Miao T, Chan JPT, Huang H, Lee PKH, Li Y. *Surface touch network structure determines bacterial contamination spread on surfaces and occupant exposure*. Journal of Hazardous Materials. 2021;416:126137.
6. **Wang P**, Zhang N, Lee PKH, Li Y. *Quantification of Lactobacillus delbrueckii subsp. bulgaricus and its applicability as a tracer for studying contamination spread on environmental surfaces*. Building and Environment. 2021;197:107869.
7. Lei H, Du S, Li C, Yung L, **Wang P**, Leung LY, Graham CA, et al. Sustained Chlorination of Hospital Surfaces Restructures the Microbiome and Virome and Diversifies Resistance Genes. *Environmental Science & Technology*. 2026.
8. Zhao P, **Wang P**, Lam T, Zhang N, Wang H, Zhang H, Liu J, et al. *High-touch surfaces with moderate contamination levels as key nodes in microbial dissemination*. Journal of Hazardous Materials. 2025:138834.
9. Jia W, Wang Q, Lung DC, Chan P-T, **Wang P**, Dung EC-H, Didik T, et al. *Co-existence of airborne SARS-CoV-2 infection and non-infection in three connected zones of a restaurant*. Journal of Hazardous Materials. 2024;480:136388.
10. Zhao P, Wang Q, **Wang P**, Xiao S, Li Y. *Influence of network structure on contaminant spreading efficiency*. Journal of Hazardous Materials. 2022;424:127511.
11. Miao T, **Wang P**, Zhang N, Li Y. *Footwear microclimate and its effects on the microbial community of the plantar skin*. Scientific Reports. 2021;11(1):20356.
12. Zhang N, Jia W, **Wang P**, Dung C-H, Zhao P, Leung K, Su B, et al. *Changes in local travel behaviour before and during the COVID-19 pandemic in Hong Kong*. Cities. 2021;112:103139.
13. Zhang N, **Wang P**, Miao T, Chan P-T, Jia W, Zhao P, Su B, et al. *Real human surface touch behavior based quantitative analysis on infection spread via fomite route in an office*. Building and Environment. 2021;191:107578.
14. Zhang N, Jia W, Lei H, **Wang P**, Zhao P, Guo Y, Dung C-H, et al. *Effects of human behavior changes during the coronavirus disease 2019 (COVID-19) pandemic on influenza spread in Hong Kong*. Clinical Infectious Diseases. 2021;73(5):e1142–e50.
15. Zhang N, Jia W, **Wang P**, King M-F, Chan P-T, Li Y. *Most self-touches are with the nondominant hand*. Scientific Reports. 2020;10(1):10457.
16. Zhang N, Su B, Chan P-T, Miao T, **Wang P**, Li Y. *Infection spread and high-resolution detection of close contact behaviors*. International Journal of Environmental Research and Public Health. 2020;17(4).

SCIENTIFIC PRESENTATIONS

Models of Infectious Disease Agent Study (MIDAS) Annual Meeting, USA

Lightning talk, 2024

Indoor Air, the 18th Conference of the International Society of Indoor Air Quality & Climate (ISIAQ), USA	Poster, 2024
MIDAS Annual Meeting, USA	Poster, 2023
College of Energy Engineering, Zhejiang University, China	Invited talk, 2023
School of Public Health, Zhejiang University, China	Invited talk, 2023
HKU–Procter & Gamble Workshop on Microbial Transmission in Building Environment, HKSAR	Moderator, invited talk, 2023
School of Public Health, Sun Yat-sen University, China	Invited talk, 2022
Indoor Air, the 17th Conference of the ISIAQ, Finland	Virtual poster, 2022
ISIAQ Healthy Buildings America, USA	Virtual oral, 2021
Indoor Air, the 16th Conference of the ISIAQ, Korea	Virtual oral, 2020
ISIAQ Healthy Buildings Asia, China	Oral, 2019
17th International Symposium on Microbial Ecology, Germany	Poster, 2018
Indoor Air, the 15th Conference of the ISIAQ, USA	Oral, 2018

PROFESSIONAL SERVICE

Member, Models of Infectious Disease Agent Study (MIDAS)	2022–present
Member, International Society of Indoor Air Quality & Climate (ISIAQ)	2018–present
Scientific and Technical Committee member, ISIAQ	
STC13–Microbes in indoor environments	2023–2024
STC31–Health effects and epidemiology	2023–2024

Journal reviewer: AJPM Focus, BMC Infectious Diseases, Building and Environment, Building Simulation, Cities, Environment International, Environmental Science & Technology, Indoor Environments, Journal of Building Engineering, PLOS Computational Biology, Progress in Biophysics and Molecular Biology, Travel Medicine and Infectious Disease

TEACHING EXPRIENCE

Teaching assistant, HKU Mechanical Engineering	
Air Conditioning and Refrigeration (MECH3429)	2017–2020
Teaching laboratory demonstrator, HKU Mechanical Engineering	
Fundamental of Electrical Engineering (MECH2406)	2016–2017

REFERENCES

Yuguo Li, Chair Professor of Building Environment, Department of Mechanical Engineering, The University of Hong Kong, HKSAR; email: liyg@hku.hk

Wan Yang, Associate Professor, Department of Epidemiology, Mailman School of Public Health, Columbia University, New York, USA; email: wy2202@columbia.edu

Patrick K. H. Lee, Professor, School of Energy and Environment, City University of Hong Kong, HKSAR; email: patrick.kh.lee@cityu.edu.hk

Andrew K. Persily, Retired, Building Energy and Environment Division, National Institute of Standards and Technology, Gaithersburg, USA; email: akp.iaqv@gmail.com